Evaluating Philadelphia's Kinship Care Expansion and Racial Disparity Reduction Efforts

Findings from AFCARS Data Exploration

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Overview

- Research Questions
- Data Sources
- Decision Trees: Variable Selection
- Comparison Counties
- County-Level Findings
- Case-level Findings
- Q & A

1. Research Questions

1. What characteristics are associated with the highest likelihood of placement?

2. Which US counties are most similar to Philadelphia?

3. What Structural Factors determine placement type?

4. Has racial disparity declined in Philadelphia's family-based placements?

5. Has Philadelphia been successful in expanding kinship care placements?

2. Data Sources

Federal Data

AFCARS

- Fiscal Years 2009 through 2023, 15 years of observations.
  - 9 million total case records.
- Over half of cases not associated with a specific jurisdiction due to anonymity rules.
- Filtered further to those 18 and under in placement on 9/30, matching the DHS point-in-time figures.

Census

- ACS 5-year demographic estimates 2009-2023
  - Race and age group population estimates

Cleaning and Matching

Not all variables are immediately usable.

- **Clinically diagnosed disability:**
  - Multiple Levels representing different trends
- **Current Placement Setting:**
  - Categorical variable, split and re-grouped into policy-relevant categories.
- **Demographics:**
  - Converted to ratios, derived population proportions.

3. Decision Tree Variable Selection

What themes influence placements?

Rationale



The AFCARS and Census includes over 200 variables.

- Too many to meaningfully assess.



Decision Trees cut through the noise while maintaining variation.



More meaningful insights.

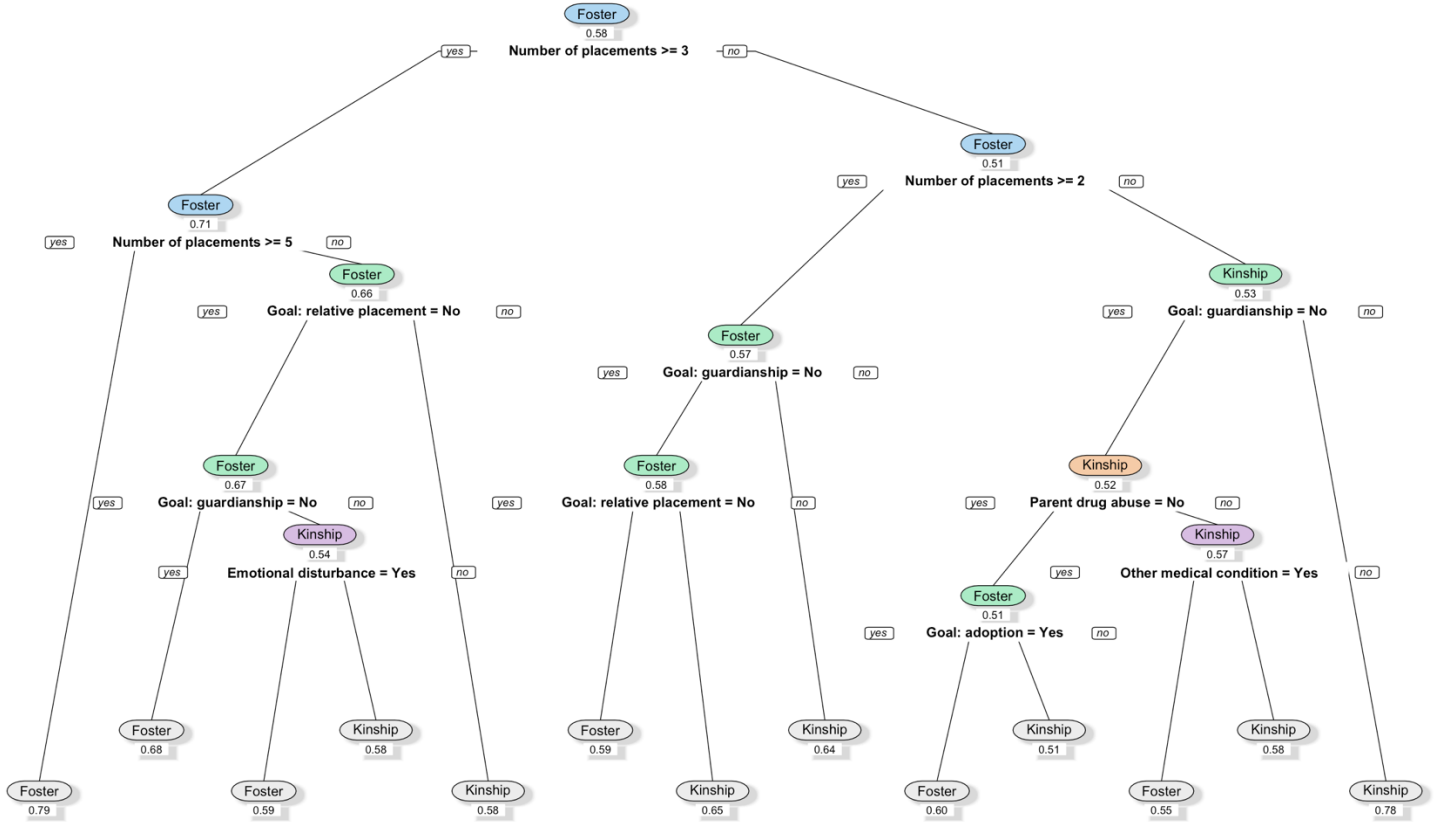
Decision Tree findings

6 overarching variable groups in the data:

Domain	Key Variables
1. Length of stay / system contact	Time since removal, lifetime LOS, avg. placements
2. System volume & county demographics	Foster count, county under-19 pop., removal rate, exit rate
3. Case goals	Reunification goal, long-term foster care goal, emancipation
4. Family structure	Single-mother caretakers
5. Child demographics (county shares)	Age at start, % Native, % White, Black overrepresentation index
6. Removal reason	Housing-related removals

Kinship Model:

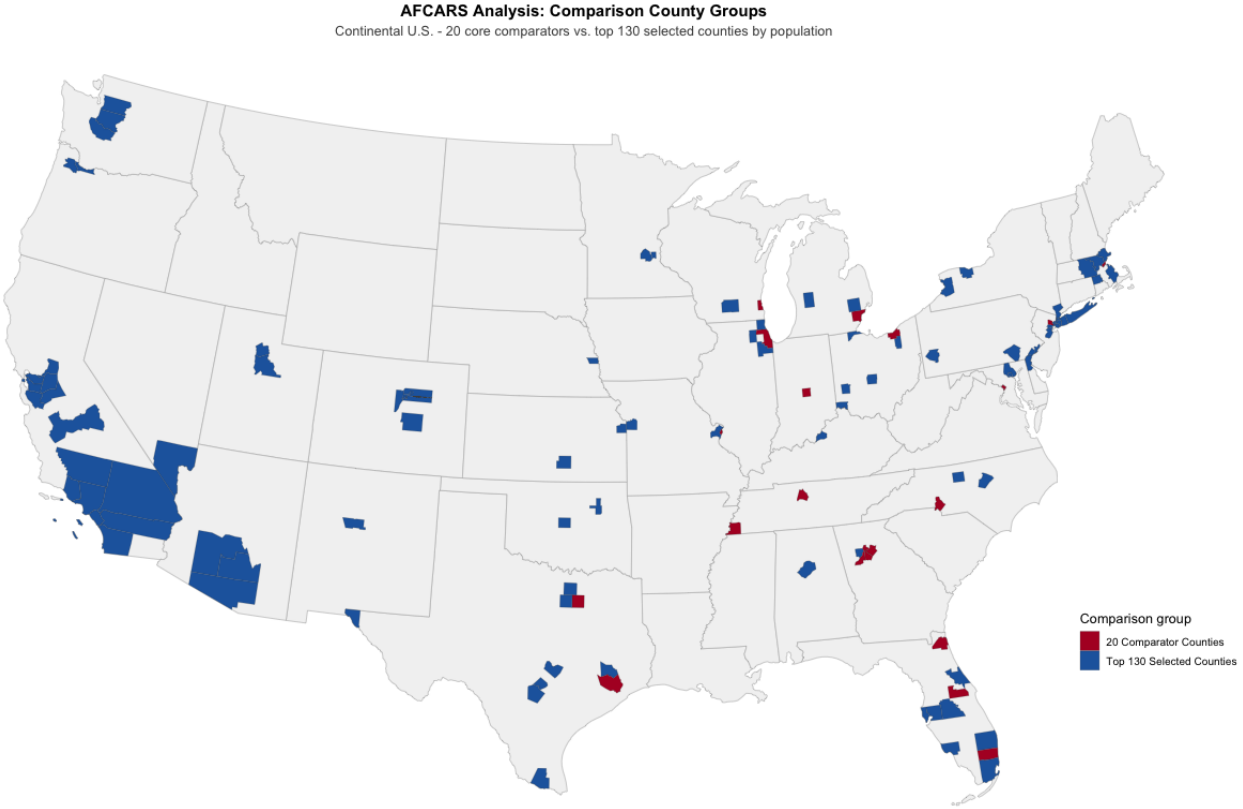
Simplified decision tree (top splits). Each box is colored by which of the 6 factor domains its question draws on.



Data: case-level AFCARS, 130 counties, 2009-2023 (kinship & foster placements; N = 1,016,383). rpart classification, depth-limited for clarity.
Boxes show the predicted placement and its probability; at each split, cases meeting the condition follow the "yes" branch.

4. Comparison Counties

Comparison Jurisdictions



Top 20 most comparable counties

Closest matches:

1. St. Louis City, MO
2. Memphis, TN
3. Newark, DE
4. Atlanta, GA
5. Detroit, MI
6. Broward, FL
7. Dallas, TX
8. DeKalb, GA
9. Milwaukee, WI
10. Jacksonville, FL
11. Indianapolis, IN
12. Cleveland, OH
13. Nashville, TN
14. Orlando, FL
15. Houston, TX
16. Charlotte, NC
17. Chicago, IL
18. Boston, MA
19. Gwinnett, GA
20. Washington D.C.

Philadelphia's Child Welfare Population is an Outlier among the most populus US counties.

Black children are overrepresented in the city's child welfare population relative to other cities. This corresponds to comparable jurisdictions in the southeast, rust belt, and mid-Atlantic regions.

Matching Process Overview

Matching over 5 variables:

1. Black/White pop. Ratio
2. Hispanic/White-NH ratio
3. Black/White under-19 pop. Ratio
4. Hispanic/White-NH under-19 pop. Ratio
5. Median household income

Distance Matching Algorithm:

Standardizes Distance across multiple variables, minimizing distance over all dimensions relative to the distribution of the dataset.

5. County-Level Analysis

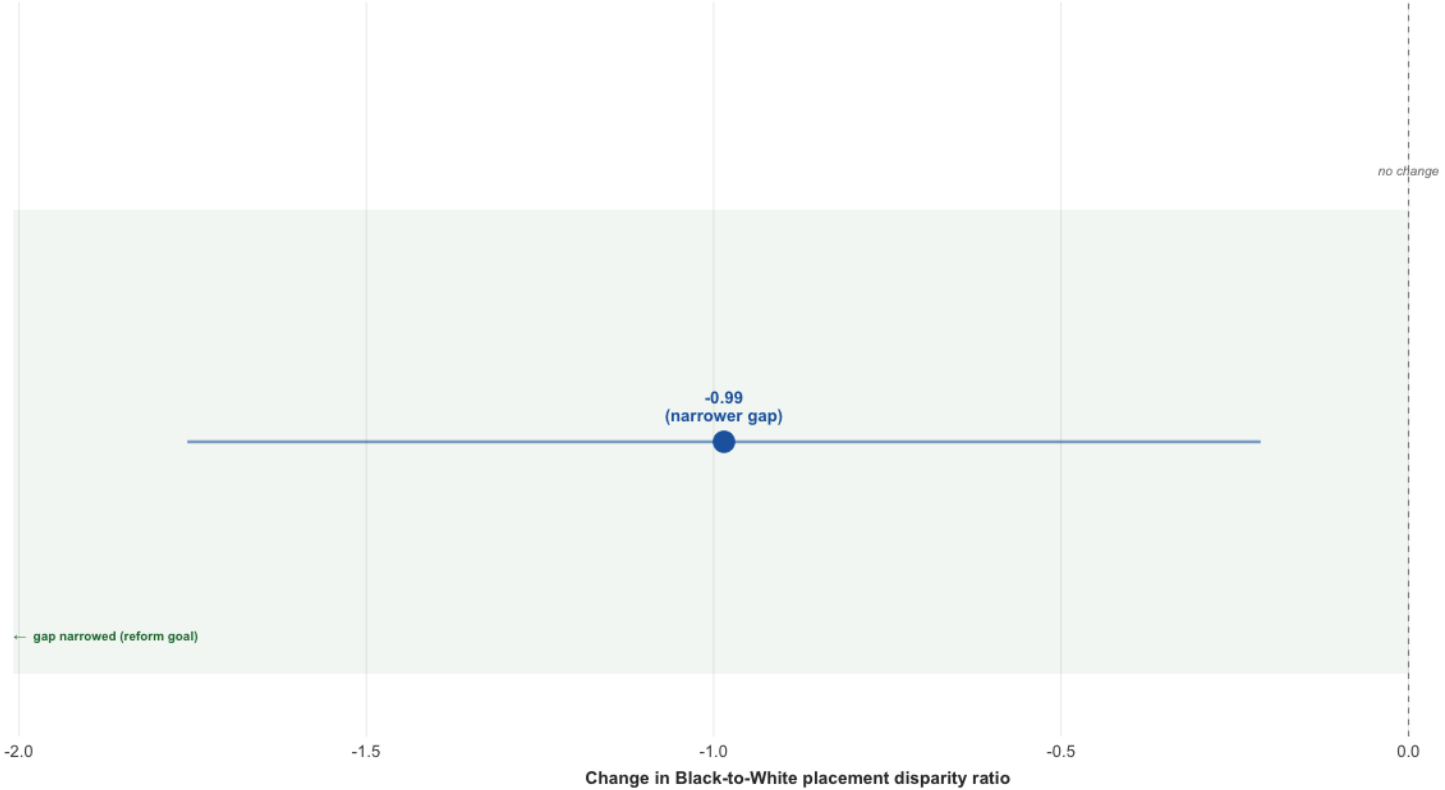
Declines in Racial Disparities

Has Philadelphia's rate of Black disproportionality in placements declined faster than comparable counties?

- Disparity ratio: $\frac{Black\ children\ in\ Placement / Under\ 19\ Black\ pop}{White\ childred\ in\ Placement / Under\ 19\ White\ pop}$
- ~ 1 point reduction in disparity ratio post-2017 in Philadelphia.
  - Above what would be expected based on comparison group.
- Two-Way Fixed Effects Differences in Differences model to isolate effects of post-2017 policies.

The Black–White Disparity in Familial Care Placements Narrowed Post-2017

Compared with 20 similar counties, Philadelphia's Black-to-White placement disparity ratio fell by about 1.0 point post-2017. This is a statistically significant improvement.



Difference-in-difference model (comprehensive spec) · 20 matched comparison counties · 296 county-year pairings · p = 0.013.
Horizontal line shows the range of statistical uncertainty (95% confidence interval).

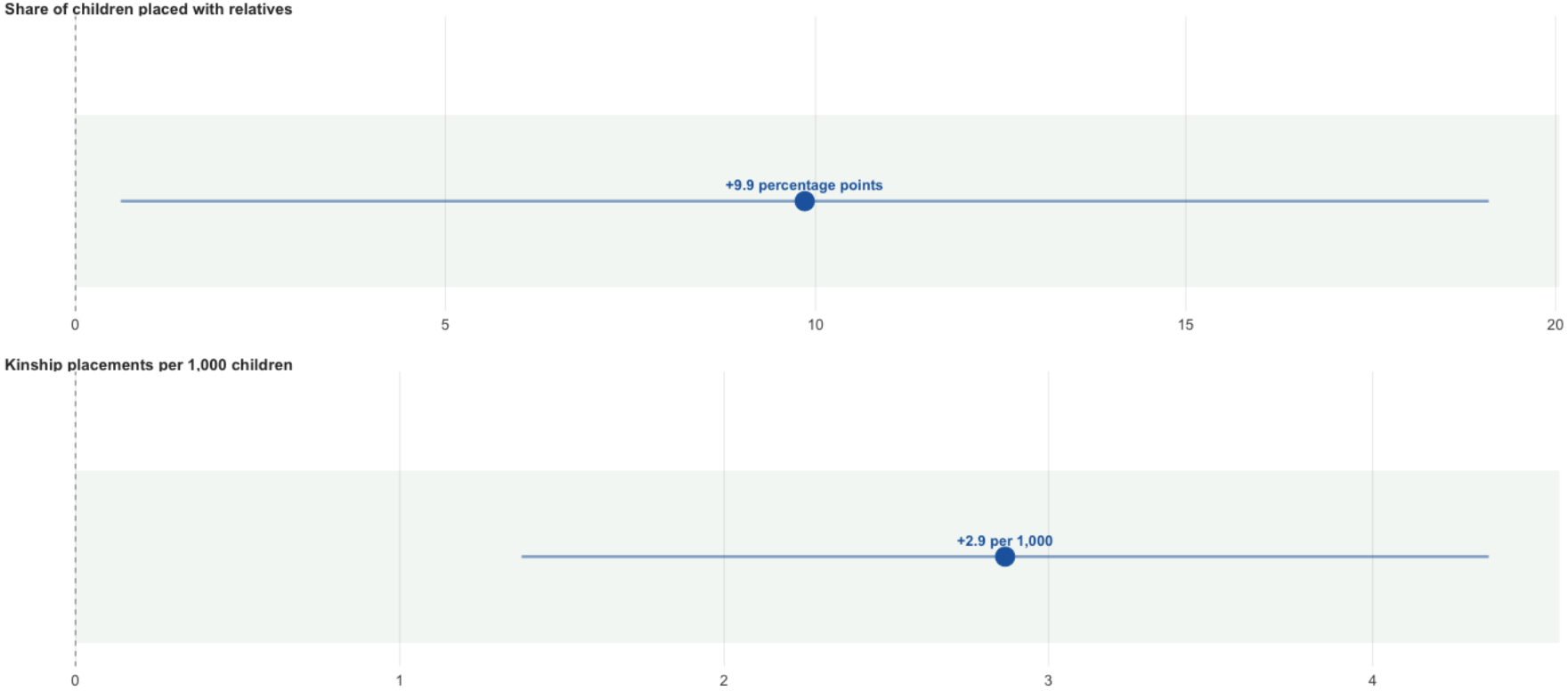
Kinship Placements

Has Philadelphia's rate of placement into kinship care increased faster than comparable counties?

- Yes, by about 9.9 percentage points above comparable counties.
- Assessed Two-way Fixed Effects differences-in-differences analysis:
  - Additional controlling variables included from decision tree analysis.
  - Isolates specific impact within Philadelphia beyond factors shown to influence kinship care rates.

Increased Placement of Children in Kinship Care Post-2017

Compared with 20 similar counties, Philadelphia placed more children with kin post-2017. Both measures rose by a statistically significant amount beyond what would otherwise be expected. Bars to the right of "no change" mean more kinship placements.



Difference-in-difference model (comprehensive spec) · 20 matched comparison counties · 296 county-year combinations.
Share: p = 0.038. Rate per 1,000: p < 0.001. Horizontal line shows the range of statistical uncertainty (95% CI).

Philadelphia Relative to Pre-2017

~ 1

Point reduction in Black overrepresentation

9.9 pp

Increase in kinship share

Structural Differences

Why are Institutional Placements higher in urban areas?

- Where a child lives and what facilities exist nearby tends to shape placement more than a child's profile.
- Rural areas under-use congregate care because beds are unavailable, **not** because youth are systematically different.

Prior Research

- Chapin Hall (2021): Group-care entry rates are 16% in urban cores vs. 10% in rural areas.
- Trend persists after controlling for child-level characteristics, including race (Chapin Hall 2015).
- Geography is the main influencing factor

6. Case-Level Analysis

Racial Disparity Reductions Through Kinship

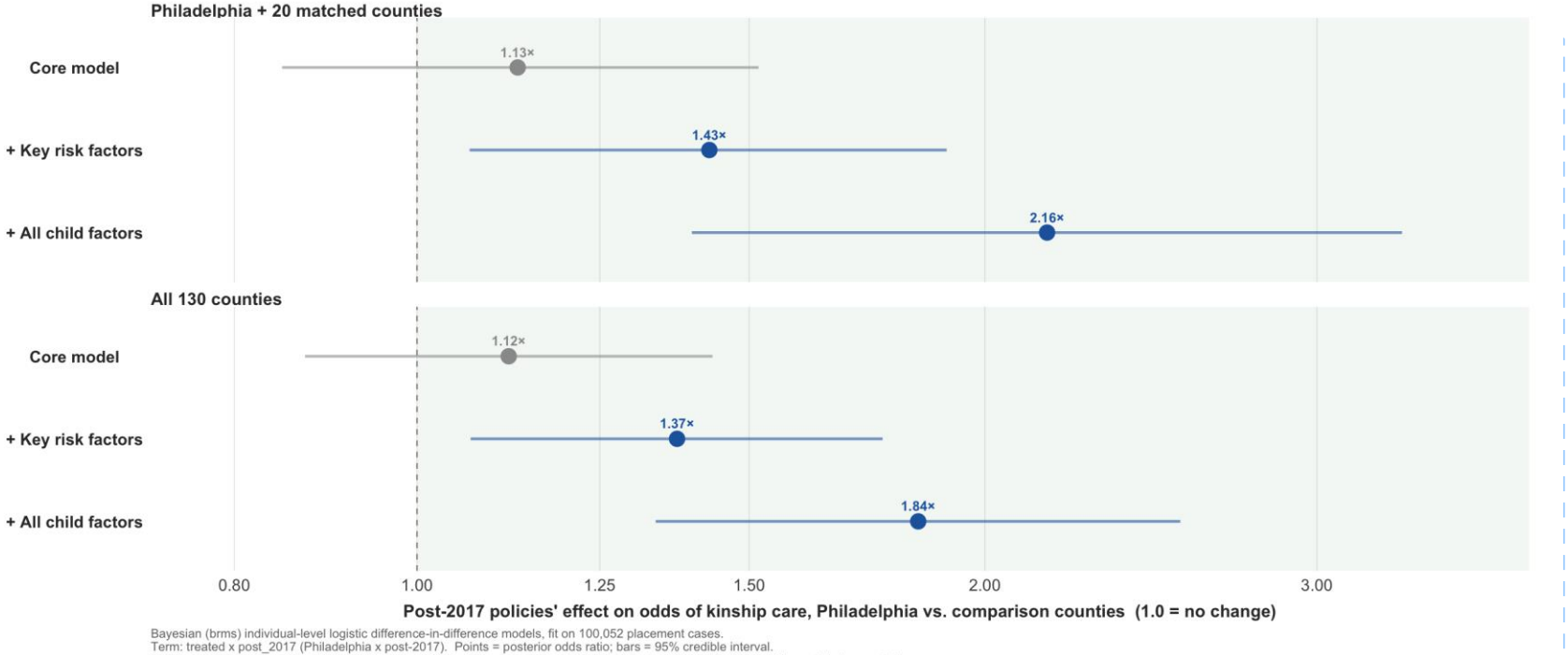
Is the increasing rate of kinship placement responsible for the observed reduction in racial disparities?

Bayesian analysis calculates specific likelihood each child enters kinship care based on their unique attributes.

- Model shows that Philadelphia children are more likely to be placed in kinship care relative to foster care.
    - Consistent with county-level findings.
- Across multiple model specifications, no specific racial effect

Philadelphia's post-2017 policies increased the likelihood of entering kinship care

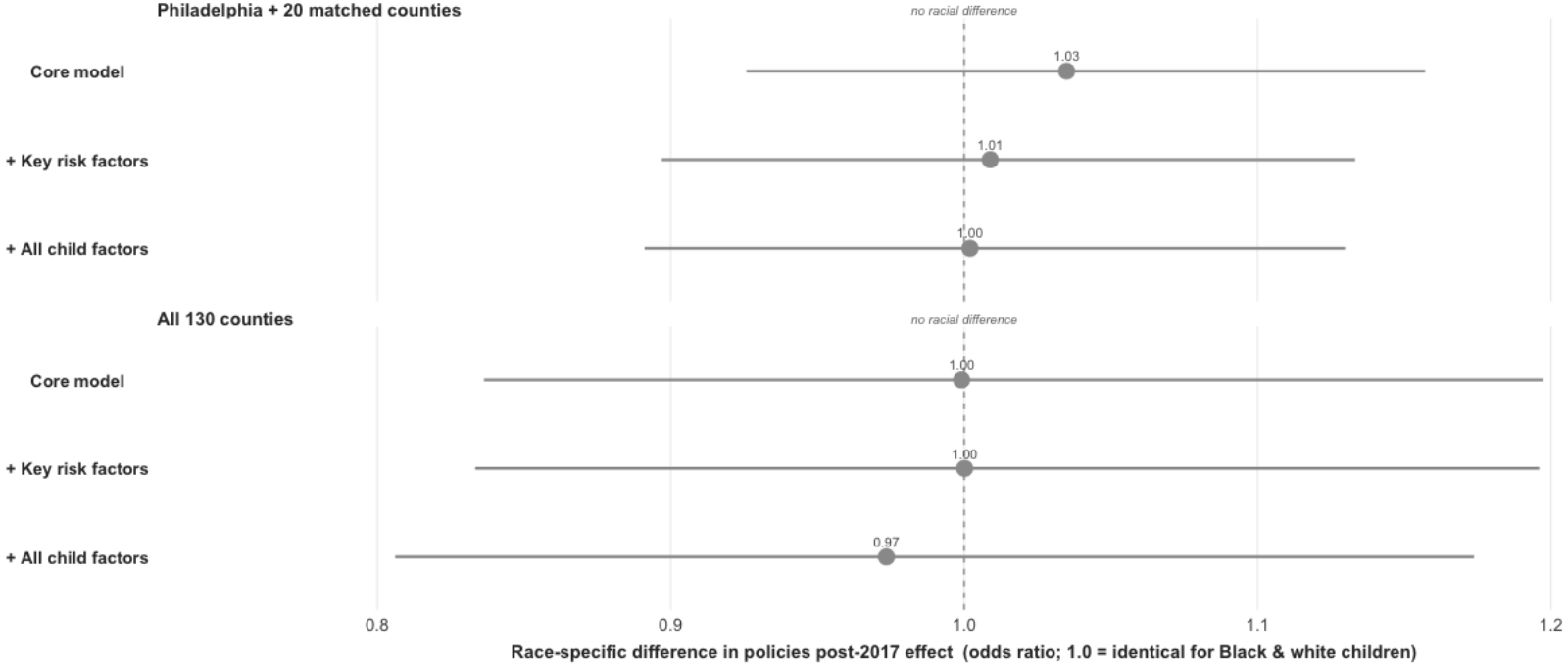
Compared with similar counties, Philadelphia's post-2017 policies are tied to higher odds of placing children in kinship care. Once a child's risk factors are accounted for, the increase is statistically significant at roughly 1.4x to 2.2x the odds. The two core models excluding controls point in the same direction but are not statistically significant.



Bayesian (brms) individual-level logistic difference-in-difference models, fit on 100,052 placement cases.
Term: treated x post_2017 (Philadelphia x post-2017). Points = posterior odds ratio; bars = 95% credible interval.
Blue = credible increase (interval entirely above 1.0); gray = increase not yet statistically clear (interval includes 1.0). x-axis is log-scaled.

Kinship placement was the same for Black and white children in Philadelphia post-2017, no racial effect

The race-specific difference in the effect (three-way interaction term) sits right at "no difference" in every model version. Every uncertainty range crosses it. Policies post-2017 neither widened nor narrowed the racial gap in who enters kinship care.



Bayesian individual-level logistic difference-in-difference models, fit on 100,052 placement cases.
Term: treated x post_2017 x black. Points = posterior odds ratio; bars = 95% credible interval (Bayesian uncertainty range).
Because all bars overlap 1.0, no racial difference in the post-2017 Philadelphia trend is detected.

Summary



Philadelphia's Efforts to increase Kinship Care are paying off

- Above increases seen in peer counties
- Other factors are driving the racial disparity decline.



Racial Disparity across placements is declining

- While caseloads nationally are declining, they are declining faster in Philadelphia for Black children.

Thank You!
Q & A

Will McCullough